

MATTEO BUTTARAZZI

Postdoctoral Researcher $\diamond$ University of Uppsala

Office: Ångströmlaboratoriet, Regementsvägen, 10, 751 06 Uppsala, Sweden. Postal address: Box 480.

✉ matteo.buttarazzi@math.uu.se  [matteobuttarazzi.github.io](https://github.com/matteobuttarazzi)  [Google Scholar](#)  [ORCID](#)

Last update: July 2026

CURRENT POSITION

Uppsala University, Uppsala, Sweden

2026–Ongoing

Department of Mathematics

Postdoctoral Research Fellow in Mathematics for Economics, Finance and Insurance

Mentor: **Erik Ekström**

RESEARCH FIELDS

My research interests include optimal stopping and free boundary problems, optimal control, problems with partial information and filtering techniques. The studied problems are motivated by financial and actuarial applications.

PAST POSITIONS

University of Turin, Turin, Italy

2025–2026

Department of Economics, Social Studies, Applied Mathematics and Statistics (ESOMAS)

Postdoctoral Research Fellow in Mathematics for Economics, Finance and Insurance

Mentor: **Tiziano De Angelis**

La Sapienza University of Rome, Rome, Italy

2024–2025

Department of Methods and Models for Economics, Territory and Finance (MEMOTEF)

Postdoctoral Research Fellow in Mathematics for Economics, Finance and Insurance

Mentor: **Claudia Ceci**

EDUCATION

La Sapienza University of Rome, Rome, Italy

2021–2024

Ph.D. in **Mathematical and Statistical Models and Methods for Economics and Finance**

Grade: Cum Laude

Thesis: **Optimal annuitization time under piecewise deterministic mortality force**

Supervisors: **Gabriele Stabile** and **Tiziano De Angelis**

Thesis defence: 03/2025

External reviewers: **Jan Palczewski** and **Enrico Biffis**

Luiss Guido Carli University, Rome, Italy

2017–2020

M.Sc. in **Economics, Institutions and Financial Markets**

Grade: 110/110 Cum Laude

Thesis: Optimal energy transition

Supervisor: **Fausto Gozzi**

VISITING POSITIONS

Uppsala University, Uppsala, Sweden

2026

Visiting Researcher (1 week). Host: **Erik Ekström**

University of Turin and Collegio Carlo Alberto , Turin, Italy Visiting Researcher (1 month). Host: Tiziano De Angelis	2024
Bielefeld University , Bielefeld, Germany Visiting PhD Student (6 months). Host: Giorgio Ferrari	2023

PUBLICATIONS

Papers in Peer-Reviewed Journals

- Buttarazzi, M., Ceci, C. (2026). DOI: **10.1142/S0219024926500032**, ArXiv: **2505.13185**.
Filtering in a hazard rate change-point model and financial applications.
International Journal of Theoretical and Applied Finance.

Conference Proceedings and Book Chapters (Peer-Reviewed)

- Buttarazzi, M., Stabile, G. (2024). DOI: **10.1007/978-3-031-64273-9_12**.
The Market Value of Optimal Annuitization and Bequest Motives.
Mathematical and Statistical Methods for Actuarial Sciences and Finance, MAF 2024, Springer.

Papers under Review

- Buttarazzi, M., De Angelis, T., Stabile, G. (Submitted in 2025). ArXiv: **2509.13091**.
Optimal Annuitization with stochastic mortality: Piecewise Deterministic Mortality Force.
- Buttarazzi, M. (Submitted in 2025). ArXiv: **2604.09342**.
Optimal Annuitization Time under a Mortality Shock.

SEMINARS AND CONFERENCES

Invited talks

- [2026] 5th Italian Meeting on Probability and Mathematical Statistics, Palermo, Italy.
- [2026] Seminar series on Stochastic Processes and Applications, Uppsala University, Sweden.
- [2025] sTOchastic control PRIN Workshop on Stochastic Control, Turin, Italy.
- [2024] XLVIII annual conference of the Italian Association for Mathematics Applied to Social and Economic Sciences (AMASES), Ischia, Italy.
- [2024] 4th Italian Meeting on Probability and Mathematical Statistics, Rome, Italy.

Contributed talks

- [2026] XXVII Workshop on Quantitative Finance, Bergamo, Italy.
- [2025] 12th General AMaMeF Conference, Verona, Italy.
- [2025] Perspectives on Actuarial Risks in Talks of Young Researchers (PARTY), Liverpool, UK.
- [2025] Second International Conference on Control and Optimization (CONTROPT), Bari, Italy.
- [2025] XXVI Workshop on Quantitative Finance, Palermo, Italy.
- [2024] XIV MEMOTEF Research Days, Rome, Italy.
- [2024] Actuarial and Financial Mathematics Conference (AFMATH), Brussels, Belgium.
- [2023] XIII MEMOTEF Research Days, Rome, Italy.

TEACHING EXPERIENCE

Lecturer

Precourse of Mathematics B.Sc. in Economics and Finance, La Sapienza University of Rome	AY 2025/2026
---	--------------

Teaching Assistant

Stochastic Calculus and Mathematical Finance

AY 2025/2026

M.Sc. in Quantitative Finance and Insurance, University of Turin

Probability for Finance

AY 2025/2026

M.Sc. in Quantitative Finance and Insurance, University of Turin

Mathematics base course

AY 2022/2023–2024/2025

B.Sc. in Economics, La Sapienza University of Rome

REFEREE

Bollettino dell'Unione Matematica Italiana (BUMI); Journal of Mathematical Economics (JME); Journal of Optimization Theory and Applications (JOTA); Mathematics and Financial Economics (MAFE).

FUNDING & GRANTS

Research Grants

Starting Research Grant¹, La Sapienza University of Rome 2024

Title: “Optimal Annuitization under Piecewise Deterministic Mortality Force”

Principal Investigator. Funding: 2,500 €. Grant No.: AR2241906EB2D3B9.

University Research Grant¹, La Sapienza University of Rome 2024

Title: “On the Role of Bequest Motives in the Annuitization Decision”

Co-Investigator. Funding: 3,000 €. Grant No.: RP124190EA4DDA9F.

University Research Grant, La Sapienza University of Rome 2023

Title: “Ambiguity: Its Role in Asset Pricing and Insurance”

Co-Investigator. Funding: 9,050 €. Grant No.: RM123188F744909D.

International Mobility Grant for PhD Students, La Sapienza University of Rome 2023

Funding: 2,100 €. Call Reference: D.R. n. 1505/2023.

University Research Grant, La Sapienza University of Rome 2022

Title: “Surrender Options in Participating Life Insurance Policies: The Impact of Penalties on Policyholders’ Decisions”

Co-Investigator. Funding: 3,000 €. Grant No.: RP12218167E24F87.

PhD Scholarship, La Sapienza University of Rome 2021–2024

Conference Grants and Support

FWO–WOG Grant, Actuarial and Financial Mathematics Conference (AFMATH) 2024

Funding: 400 € and full registration fee waiver.

Registration Fee Waiver, 4th Italian Meeting on Probability and Mathematical Statistics 2024

Approximately 70% registration fee waiver.

INTERNATIONAL AND NATIONAL RESEARCH GROUPS

Member, PRIN 2022 – European Union Next Generation EU Project 2024–ongoing

Project title: “Stochastic Control and Games and the Role of Information.”

Grant code: BEMMLZ

¹Competitive university-wide grants across departments and schools supporting early-career researchers (Starting Research Grants) and research projects (University Research Grants) ([official call](#)).

MEMBERSHIPS

Association for Mathematics Applied to Social and Economic Sciences (AMASES) 2024–ongoing

SUMMER SCHOOLS

[2023] 32nd Jyväskylä Summer School, Jyväskylän Yliopisto, Finland

[2023] Risk and Uncertainty in Economics, Insurance and Finance, Bielefeld, Germany

[2022] Summer School SMI, Perugia, Italy

NON ACADEMIC EXPERIENCE

Credit Analyst at Younited Financial S.A. 2020-2021